



Mark Scheme (Results)

June 2025

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H/1B)
Unit 1: Principles and Applications of Science I
(Biology)

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Unit 1: Principles and Applications of Science I (Biology)

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark															
1(a)(i)	nucleolus	reject nucleus accept chromosomes, DNA	1															
1(a)(ii)	<table><tr><td>organelles</td><td>found in an animal cell</td><td>found in a plant cell</td></tr><tr><td>Golgi apparatus</td><td>✓</td><td>✓</td></tr><tr><td>chloroplasts</td><td></td><td>✓</td></tr><tr><td>plasmodesmata</td><td></td><td>✓</td></tr><tr><td>ribosomes</td><td>✓</td><td>✓</td></tr></table> one mark per correct row.	organelles	found in an animal cell	found in a plant cell	Golgi apparatus	✓	✓	chloroplasts		✓	plasmodesmata		✓	ribosomes	✓	✓		3
organelles	found in an animal cell	found in a plant cell																
Golgi apparatus	✓	✓																
chloroplasts		✓																
plasmodesmata		✓																
ribosomes	✓	✓																
1(b)	C size of image size of actual object		1															
Total = 5 marks																		

Question Number	Answer	Additional Guidance	Mark
2(a)	B part B		1
2(b)(i)	Any one from: fat / fatty material / lipid cholesterol / LDL	ignore smoking ignore alcohol accept calcium accept salt (sodium) accept sugars	1
2(b)(ii)	 difference (1) 4.2 – 2.9 substitution (1) $1.3 \div 4.2 \times 100$ evaluation (1) 30.95 (%) OR substitution (1) $2.9 \div 4.2 \times 100$ difference from 100 (1) $100 - 69.05$ evaluation (1) 30.95 (%)	30.952380952 (%) correctly rounded with no working gains maximum marks	3

2(b)(iii)	Any three from: white blood cells migrate to the site of (artery wall) damage (1) (white blood cells can become) foam cells (1) by engulfing {cholesterol / lipids} (1) (foam cells) trigger programmed cell death (1) cascade effect in the inflammatory response (1)	accept named example of white blood cell, e.g. leukocytes / macrophages / monocytes accept LDL or HDL accept fat accept apoptosis	3
Total = 8 marks			

Question Number	Answer	Additional Guidance	Mark
3(a)	<u>inner</u> membrane / cristae		1
3(b)	ATP binds to myosin (1) to {slide / move} myosin over actin (1) to detach myosin from actin / ATP hydrolysis returns myosin to its resting position (1)	accept thick filament for myosin accept thin filament for actin accept crossbridge formation between myosin / actin accept enabling the power stroke between myosin and actin	2
3(c)	(slow-twitch muscle fibres) have little glycogen present (1) (many capillaries) give high concentration of {oxygen / glucose} (1) (this enables) aerobic respiration / prevents build-up of lactic acid (1)	ORA ignore fatigue	3
Total = 6 marks			

Question Number	Answer	Additional Guidance	Mark
4(a)	phagosome	accept (pathogen is) engulfed / in a vesicle accept lysosome fuses with vesicle	1
4(b)	(at the first exposure) {a low concentration of antibodies / antibodies produced slowly} due to the unrecognised (antigens on the) pathogen (1) (some) B-cells differentiate into memory cells (and remain in the blood) (1) (at the secondary exposure) the memory cells rapidly {proliferate / differentiate} into plasma cells (1) (these plasma cells then) respond by rapidly producing (large quantities of) antibody (1)		4
Total = 5 marks			

Question Number	Indicative Content
5	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content, but learners should be rewarded for other relevant answers.</i> <u>cell theory</u> • cells are the basic units of organisms • produced from existing cells during cell division, which is part of growth • required for the structure and functioning of all living organisms • similar cell types group together to form tissues that have a specific function • different tissue types group together to form organs and organ systems that have a physiological function <u>skin</u> • the skin has different layers made up of different tissues • formed by cell division and cell differentiation during growth • involves communication between different cells • contains blood, sweat glands and hair • skin has a specific function • tissues are organised to perform specific physiological functions • skin is an organ <u>blood</u> • blood contains different types of cells • including red and white blood cells • involved in the transport of oxygen and the immune response • the cells work together with a specific function • blood is a tissue

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Adequate interpretation, analysis and / or evaluation of the scientific information with generalised comments being made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The explanation shows some structure and coherence.
Level 2	3–4	• Good analysis, interpretation and / or evaluation of the scientific information. • Lines of argument mostly supported through the application of relevant evidence. • The explanation shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Comprehensive analysis, interpretation and / or evaluation of all pieces of scientific information. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The explanation shows a well-developed structure which is clear, coherent and logical.
Total = 6 marks		



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